Can AI Agents Be Organized More Efficiently for Discovering Science?

Haochuan Kevin Wang [1] **Ruijie Shi** [1]

Abstract

Teams of language-model agents add cost; when do they add accuracy? A budget can buy more calls whose answers are combined (*width*), more reasoning or a stronger model per call (*depth*), or a check that picks among answers (*verification*). We measured each at matched cost with Claude and GPT models, under preregistration and hard spending caps. A 19-agent Haiku team did not clearly beat one Sonnet call (62.5% against 71.3% on 80 BBEH questions; difference +8.8 points, −6.3 to +22.5). The reason is shared failure: six proposers were worth 1.9 independent answers, and on public leaderboards item difficulty accounts for 0.73 of the latent variation in success on SWE-bench Verified; a factor model with one difficulty per item prices the tail that all held-out systems fail (5.9% predicted, 7.6% observed). At matched cost, one reasoning call beat a vote of quick copies (confirmatory; 3.7 and 10.7 points). Verification selects but cannot create: critics and refiners recovered none of the 20 questions that no proposer answered, and a judge is capped by its candidates' coverage. Different members helped only where the best member left room: on LAB-Bench biology a team of Haiku and Sonnet gained +6.3 points over Sonnet alone at 4.4× its cost, no GPT team beat GPT-6.1 Sol alone, and a preregistered panel mixing the providers did not beat GPT's own vote (−0.7, −4.9 to +3.5). Smarter coordination rules did not pay: a learned rule for when to ask a teammate failed its preregistered test, and a recovery rule for a lost team member (PHOENIX, 330 sealed recoveries on physics and ecology tables) showed no resolved gain where it could be isolated (+0.20, −0.31 to +0.95); a stronger planner model drove the gains. Which members are called matters more than how they are organized: every winning cell of our budget map is a single model. Public-data results are post hoc, and every number carries its status.

[1]Massachusetts Institute of Technology, Cambridge, MA, USA. Correspondence to: Haochuan Kevin Wang.

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1. Introduction

Suppose an agent system has a fixed budget per task. It can ask a cheap model several times and vote, ask a stronger or longer-reasoning model once, or ask several models and let a judge choose. Which should it buy first? The answer depends on one quantity: how much the systems' failures are *shared*. If every system fails the same hard items, extra calls fail with them, and only a system that moves the hard items helps.

Six questions, in the order we met them. (1) Can a team be compressed? A pruned 19-agent team was matched by one stronger call (Section 3). (2) Why didn't 19 agents beat one call? Failures are dominated by shared item difficulty, so width is capped by the items everyone fails (Sections 3 and 4). (3) If copies don't help, what does a dollar buy? Reasoning moves a system along the same difficulty axis, and the first effort step of cheap models beat a dollar-matched vote in a preregistered test; verification selects among answers someone produced and is capped by their coverage (Sections 5 and 6). (4) Do different members help where copies don't? Only where the best member leaves room (Section 7). (5) Does a smarter coordination rule help? Neither a learned rule for asking a teammate nor a recovery rule showed a gain we can credit to the rule; the stronger member did (Section 7). (6) So the budget question reduces to choosing members (Section 8).

Contributions.

- **The Atlas**: one factor model fitted to 173 SWE-bench Verified submissions, 126 MathArena configurations over 23 competitions and 165 LiveBench models, with intervals for every systematic share and held-out tests of what it predicts about sets of systems: coverage, the all-wrong tail and vote accuracy.
- **Depth on public data**: 19 pairs of one model at two adjacent reasoning settings, pooled with random ef-

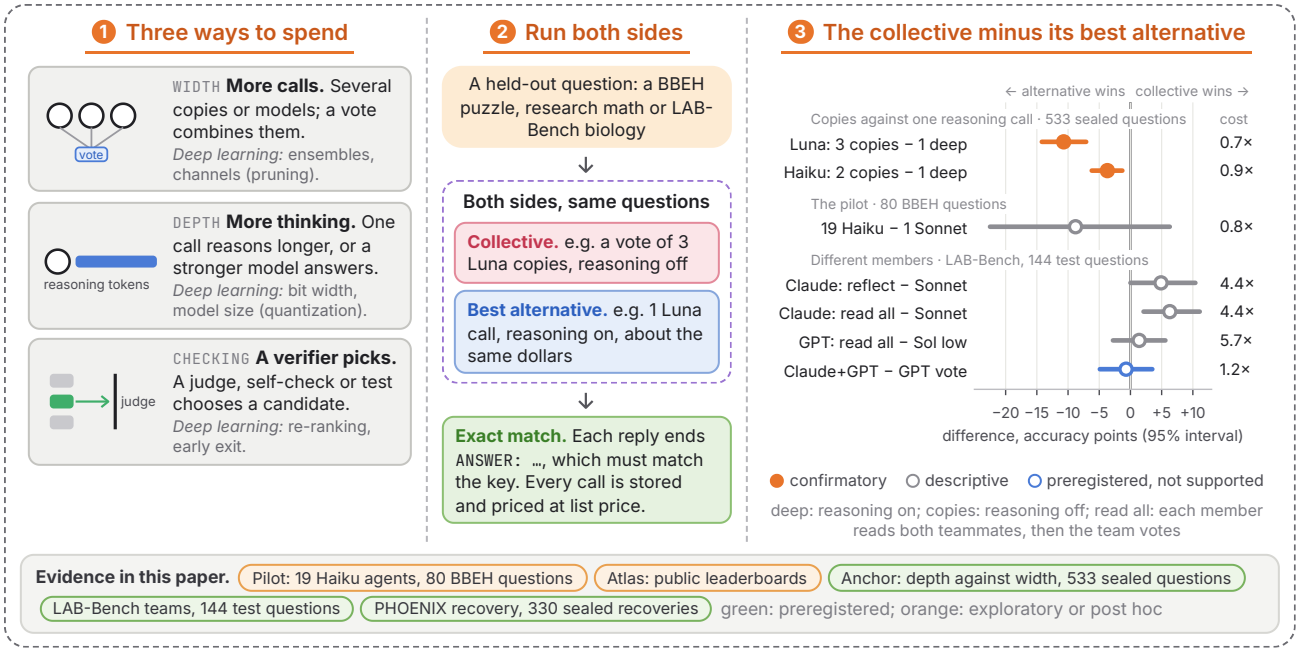


Figure 1. How every comparison in this paper works, and what it found. (1) A budget can buy width (more calls whose answers are combined), depth (more reasoning per call, or a stronger model) or checking (a verifier choosing among candidates); each has a deep-learning counterpart. (2) A collective and its best single alternative answer the same held-out questions, every reply is graded by exact match with the answer key, and every call is stored and priced. (3) The collective minus its best alternative, in accuracy points with 95% paired-bootstrap intervals; left of zero the alternative wins, and the right column is the collective's cost as a multiple of the alternative's. Rows 1–2 (CONFIRMATORY; 533 sealed questions): a vote of reasoning-off copies against one reasoning-on call of the same model at about the same dollars. Row 3: the pilot's 19-agent Haiku team against one Sonnet call (80 BBEH questions). Rows 4–7 (LAB-Bench, 144 test questions): Claude's team with every member thinking again, or reading both teammates, against Sonnet low alone; GPT's team against Sol low alone; and a preregistered Claude + GPT panel against GPT's own vote. The box below lists the studies; green marks those with preregistered tests.

fects, and a dollar-matched comparison of one deeper call against a wider vote (post hoc).

- **Preregistered API studies**: a Phase 1 case where one reasoning call wins, and an anchor study on 533 sealed items with two confirmatory tests (AH2: the first effort step against the nearest-dollar vote; AH3: the factor model's vote prediction), read once with frozen code.
- **Team studies**: a 19-agent pilot; LAB-Bench teams for each provider with a preregistered rule for when to ask a teammate and a preregistered cross-provider panel; and PHOENIX, a recovery rule for a lost member, on 18 cases of three science tables with every predictor sealed before scoring.
- **A budget map**: the best team for each dollar and latency budget, chosen from stored calls at no cost.
- **Process**: sealed items, single reads, preregistered stop rules and 135 independent audits, for $196.64 of API spend in total. Every number in this paper is generated from a committed file and carries a status (Appendix A).

2. Definitions

We define each term in words first. **Width**: more calls of the same setting, or of different systems, combined by a rule (a plurality vote) or a selector. **Depth**: more reasoning per call (a higher effort or thinking budget) or a stronger tier. **Verification**: choosing among candidate answers, by a judge model, a self-check, execution, or a label-free rule such as Dawid–Skene (Dawid & Skene, 1979).

The factor model. System s answers item i correctly with probability $P_{si} = \sigma(a_s - \beta_s b_i)$, $\sigma(x) = 1/(1+e^{-x})$. The *ability* a_s is how good the system is overall; the *difficulty* b_i is how hard the item is for everyone, the shared part of failure (scaled to mean 0 and standard deviation 1); the *failure slope* β_s is how sharply the system's success falls with difficulty. Level L0 fixes $\beta_s = 1$. A *family shock* $u_{f(s),i}$ adds a shared draw for systems of one vendor; *call luck* is what remains, identified only from repeated calls. Fits are penalised logistic regressions (ridge 0.1), so items every system gets right or wrong keep finite difficulties.

Derived quantities. The *systematic share* $\mathrm{var}(b)/(\mathrm{var}(b) + \pi^2/3)$ is the share of latent variation in success that items share ($\pi^2/3$ is the variance of the logistic call-level noise). *Coverage* of a set of systems is the share of items at least one member gets right: the ceiling of any rule that returns one member's answer, as Chen (2026) also show; the *all-wrong rate* is one minus coverage. The *vote ceiling* of one system is the share of items whose most common answer is right, the limit of infinitely many repeats. *Selection efficiency* $\eta = (\text{selected} - \text{best single})/(\text{coverage} - \text{best single})$ is the share of the room above the best member that a selector reaches. A *consensus inversion* is a label-free rule rating a weaker system above the best one. *Calibration by dilution*: adding near-random votes shrinks a panel's calibration gap without improving its ranking (F13 in our findings ledger).

Statuses. CONFIRMATORY: a preregistered test in a Holm family, read once on sealed items. PREREGISTERED-DESCRIPTIVE: a preregistered quantity, read once, no pass/fail. EXPLORATORY: on sealed items, not preregistered as a test. POST HOC: chosen after the data were read, or on data not sealed for it; all public-data analyses are post hoc in this sense, as are numbers computed after a read. Post hoc results are many and uncorrected for multiplicity; we read them as descriptions, not tests.

3. The pilot: nineteen agents and one call

Our first study pruned a team rather than measuring a lever (Figure 1, row 3). Six Claude Haiku 5.5 proposers answered each question, six critics read all six answers, six refiners revised, and one aggregator answered: 19 calls. On 80 BBEH questions the full team scored 62.5% at $0.0115 per question; one Sonnet 5.5 call with thinking off scored 71.3% at $0.0144, a difference of +8.8 points (−6.3 to +22.5; EXPLORATORY). The pruned teams tied within this noise.

The proposers' right-or-wrong records correlate at 0.43 on average, so six are worth 1.9 independent answers (effective size $n/(1+(n-1)\rho)$, a measure of redundancy rather than a number of agents one could deploy). A plurality vote of one proposer scores 44.6% and of all six 52.1% (averaged over every set of proposers), while the chance that at least one is right rises from 44.6% to 75.0%; independent errors at the same accuracy would give 97%. Accuracy rose through critique (57.5%) and refinement (60.2%) to 62.5%: on the 19 questions where one or two proposers were right the team reached 73.7%, and on the 20 where none was right it never recovered (0.0%). Checking selected among answers someone produced; it did not create them. The stages were not ablated one by one. The next two sections ask why failures are shared and what a dollar buys instead (Figure 2).

4. The Atlas: what public data say about width and depth

Data. SWE-bench Verified (Jimenez et al., 2024): 173 submissions on 500 tasks, one run each, 39 with per-task cost and step counts (mini-SWE-agent). MathArena (Balunović et al., 2025): 126 configurations on 733 final-answer problems from 23 competitions, usually four calls each, with answers grouped into classes by the grader. LiveBench (White et al., 2024): 165 models on 228 items of three binary tasks, one call each, without answers or costs. Each benchmark is fitted on its own. All Atlas numbers are POST HOC.

4.1. Failures are mostly shared

The systematic share is 0.73 on SWE-bench Verified (item bootstrap 0.70–0.75; over 40 random halves of the systems, 0.71–0.73), 0.73 on MathArena (0.72–0.75) and 0.65 on LiveBench (0.61–0.68); our Phase 1 BBEH panel of four arms sits lower at 0.42 (0.38–0.47; Appendix Figure 6). On the 39 SWE-bench submissions with per-task data, difficulty alone implies 98% of the mean pairwise failure correlation (0.502 of 0.512); two systems of one company add only 0.026 of residual correlation. All 39 fail 8.6% of tasks, against 0.0% if they failed independently.

4.2. Width is capped by the shared tail

Coverage and the all-wrong tail. Fitting difficulties on half the systems and each held-out system's ability on half the tasks, the model predicts the coverage of held-out 2–5 system sets on held-out SWE-bench tasks within 1.5–1.8 points; independence misses by 11–20. Chen (2026) report that a calibrated one-factor copula underprices the all-wrong tail by about 2.5× on open-ended mathematics. We give the factor model and the copula the same information, the calibration tasks only: the factor model draws an unknown test task's difficulty from the calibration tasks' difficulties, the copula fits its correlation to the calibration tasks' pairwise failures. For all 87 held-out SWE-bench systems on 250 held-out tasks the observed all-wrong rate is 7.6% (Clopper–Pearson 4.6–11.6); the factor model predicts 5.9% (observed over predicted 1.29, 0.68–1.84), the copula 3.5% (2.15×) and independence 0.00% (Figure 9). The factor model's free distribution of difficulties carries the tail that a Gaussian latent factor misses. On the small MathArena sets (15–26 test problems) both models underpredict the 5-system tail, on 2 and 2 of 6 sets; the math tail is unresolved.

Votes of one system. A vote of two calls cannot gain in expectation (a right–wrong split is a tie), so gains start at three. The factor model's *vote rule* predicts a k-vote from one-call success p, the share z of failures that abstain (abstentions are non-votes) and the chance q that two wrong

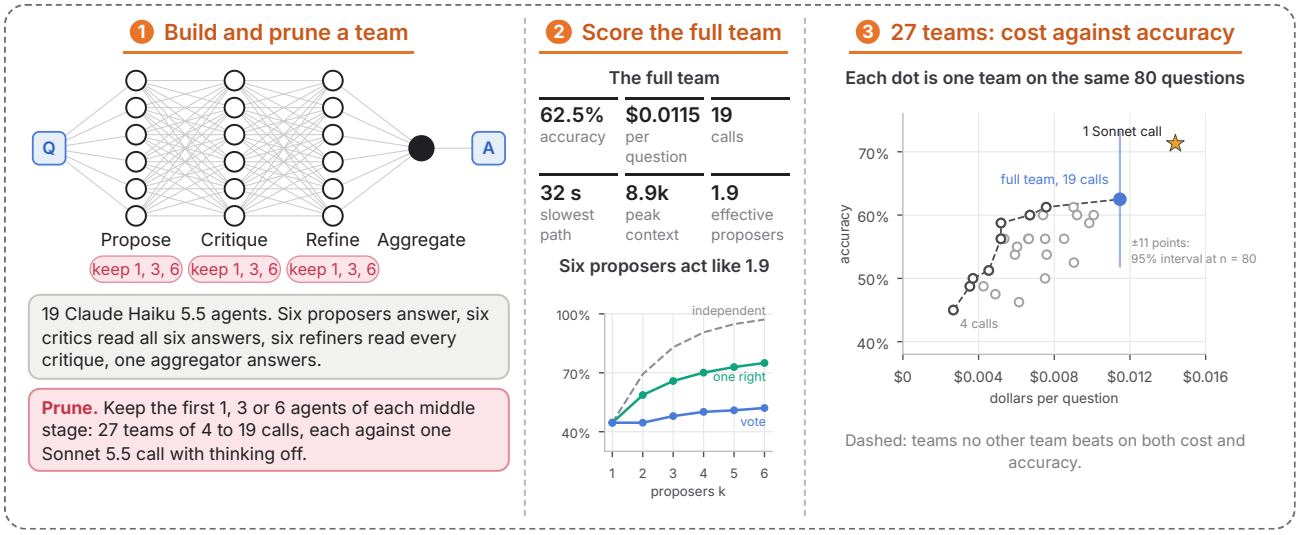


Figure 2. The pilot. (1) Nineteen Claude Haiku 5.5 agents in four stages; pruning keeps the first 1, 3 or 6 agents of each middle stage, which gives 27 teams. (2) The full team's measurements, and why it does not gain more: a plurality vote of k proposers (averaged over every set of k) against the share of questions at least one of the k answered right, and that share if their errors were independent; six proposers are worth 1.9 independent answers. (3) All 27 teams on the same 80 BBEH questions, accuracy against dollars per question; the vertical bar is the full team's 95% interval, wide enough to cover most teams, and the star is one Sonnet 5.5 call with thinking off (EXPLORATORY).

answers coincide. Cross-fitted on 695 (system, set) cells with three calls in 24 sets, its mean absolute vote-gain error is 2.0 points against 2.2 for the zero-gain predictor "votes add nothing"; predicted and observed gains correlate at 0.46, and the gain's squared error is smaller by 2.4 points^2 (0.6 to 3.4; item noise adds the same variance to both and cancels). But the rule overstates gains (4.7 predicted, 2.9 observed where it predicts 3 or more), and on the 3-vote accuracy itself it is worse than zero-gain (mean absolute error 4.4 against 4.0 points; squared error larger by -4.2 points^2, -9.1 to -1.6). The rule ranks where votes pay; it does not yet price them.

Votes across systems. Votes pay most when weak systems with different strengths are mixed. On 1,577 panels of three and 1,585 of five MathArena systems with one-call accuracy 0.2–0.7, the plurality of one call each gains 5.7 and 12.5 points over a random member. Cross-fitted, the factor model predicts the panel vote with bias 0.4 and 0.4 points and absolute error 6.4 and 6.6, against 8.2 and 13.6 for zero-gain and 11.9 and 16.2 for independence; it is closer than zero-gain in 4 and 4 of 5 sets, and closer than independence in 4 and 5. Where composition matters, the model prices it without bias; independence overstates it and zero-gain ignores it.

4.3. Depth moves ability, not the failure pattern

For each of 19 public pairs of one model at two adjacent reasoning settings (11 models, some on two benchmarks; difficulties from every other system), we fit both settings' ability and slope on the items both answered (Figure 10). Pooled with random effects, ability rises by 0.92 logits (0.59–1.24; between-pair variance 0.35); the slope changes by 0.03 (-0.14–0.21; between-pair variance 0.00). One pair per model and benchmark (13 pairs) gives 1.21 and 0.11. Held out, the higher setting's calls are predicted better by the lower setting plus one shift than by a refit that may also change the slope (log loss per call 0.0041 in favour of the shift, 0.0022–0.0061) and far better than by its base rate (0.192 nats per call). The difficulties are the population's, so this says the higher setting fails the items that are hard for everyone; the lower setting's own item outcomes plus a shift predict no better (0.007, -0.017 to 0.030). Two pairs left out after a look at output tokens (one setting of unknown order, one different checkpoint) are shown in the extension report and were never pooled.

Depth against width at equal dollars (post hoc). For 10 MathArena pairs whose lower setting has three or more calls, one call at the higher setting minus the vote of the lower setting with the nearest dollars per item (the 3-vote in every pair) is 12.7 points (7.1–18.2); on the 7 pairs priced within a factor of 1.5 it is 11.9 (6.0–17.7; Appendix Figure 7). The step pays where a model starts low and about nothing at the top of its range: +1.1 points for Gemini 3.1 Pro medium to default and -1.5 for Opus 5.5 high to max. We logged, before it is read, that this predicts about zero for the preregistered CD2 test of a top Haiku effort step. Price matters as much: on SWE-bench Verified MiniMax M2.5

(high) resolves 75.8% at $0.073 per task against 76.8% at $0.754 for Claude Opus 4.5 (high), 10× the price for one point (Figure 9, right).

5. Preregistered API studies

Phase 1: the saturated case. On 300 sealed BBEH items (Kazemi et al., 2025), one GPT-6.1 Sol call at low reasoning was right 97.3% of the time at $0.0062 per call; Claude Sonnet 5.5 without reasoning 72.0% at $0.0139 (PREREGISTERED-DESCRIPTIVE). Width bought little: 5 Haiku calls are worth 1.84 independent calls (repeat correlation 0.43), their plurality reached 48.3%, a 12-call mixed panel 58.0%, and none of 276 cascade settings came within 2 points of Sol at 0.8× its dollars (EXPLORATORY/POST HOC). The confirmatory cascade test (H7) passed non-inferiority to Sonnet (−0.7 points) only by escalating 78% of items. Sol's accuracy is not answer memory: 40 of 40 perturbed items right. The Sol–Sonnet gap mixed vendor and reasoning, which motivated the anchor.

The anchor. 440 BBEH items (linguistic puzzles, cartoon captions, word sorting, causal judgement) and 93 arXiv-Math problems were sealed before any call. Arms: Claude Haiku 5.5 (off, low, high), Sonnet 5.5 (off, low), GPT-6 Luna (none, low, medium), GPT-6.1 Sol (low, medium), most with three calls per item; dollars are measured at list prices. Each block was read once with frozen code; Holm at 5% over the two tests that kept power of at least 0.70 (AH2, AH3).

*AH2 (*CONFIRMATORY*, passes; $p = 0.001$).* Raising a cheap model's reasoning one step beat repeating its no-reasoning call at the nearest measured dollars: Haiku at effort low against the Haiku-off 2-vote by 3.7 points (1.3–6.2), Luna at reasoning low against the Luna-none 3-vote by 10.7 (7.1–14.2); the deeper call cost 1.17× and 1.38× as much. For Haiku the 2-vote equals one call in expectation, so this shows that effort low beats effort off at slightly more dollars, not that depth beats a wider vote; against the dearer 3-vote the difference is 1.8 (−0.7–4.3), not tested. The effect sits mainly in word sorting (and linguistic puzzles for Luna; computed after the read). The decision as frozen: for cheap models on these items, buy the first effort step before width. It covers neither higher steps nor other tasks.

*AH3, the vote rule (*CONFIRMATORY*, passes).* Cross-fitted on single calls, the factor model predicted the six 3-call arms' 3-vote accuracy with mean absolute error 0.8 points (upper bound 2.4) and their gain over one call within 0.7 (upper bound 1.3). The gains were small (0.4 to 2.6 points) and zero-gain also meets the limits (gain error 1.5 against 0.8), so AH3 shows accuracy where votes gain little; P1 above finds the same rule overstating larger gains.

*AH1 and AH4 (*EXPLORATORY*).* Sonnet's effort-low setting closed none of the 11.1-point gap to Sol low ($R = 0.02$, −0.32 to 0.26), so the vendor–reasoning confound stays open. The Sol judge's selection efficiency was 0.85 on easy items and 0.14 on hard ones (difference 0.71, 0.41–1.05), the predicted direction, with a ratio whose hard-half denominator is small.

The depth study. A preregistered study of Haiku at every reasoning effort (research/v4/cd/read.md) finds that accuracy gains flatten by the second-highest effort; the drop at the highest effort is within its interval and coincides with calls cut at the output limit.

6. Verification

Label-free selection. With repeats, Dawid–Skene rated Sol at 62.5% on Phase 1 against a true 97.3% (EXPLORATORY); on the anchor its pooled variant put Sonnet at 79% against a true 45%, and its answers (47.5%) stayed below one Sol low call (56.2%). A simulation map (Figure 11) shows when this happens: with one strong system among weak ones and one call each, plain Dawid–Skene ranks a weak system first in 90% of panels; a fixed label space removes the inversion in the simulation (0%) but still loses 8.0 points to the strong system. On MathArena the fixed label space does not help: beside a lone strong system it ranks another system first in 62–71% of panels, against 42–63% for plain Dawid–Skene. At matched calls (three systems with one call each, against the best system's own 3-vote, the best chosen with labels) every policy loses beside a lone strong system (plurality −34.0 points, plain Dawid–Skene −25.7) and none gains among competent ones (plurality −1.8, plain Dawid–Skene −1.0, fixed labels −0.3).

The anchor's judge, and why it cannot test judges. On the anchor's BBEH items the Sol low judge, choosing among the answers of two cheap pools, reached 52.3%: 13.3 points above the best pool arm (9.8–15.8), selection efficiency 0.70 (0.57–0.80). Sol low answering alone scored 60.6%, so the judge sat 8.3 points below it (−12.0 to −4.8) at 1.6× the dollars (Appendix Figure 8; POST HOC, computed after the read). But the pools covered only 58.0% of the items: even a perfect selector over them would have lost to Sol alone. The comparison measures the pools, not the judge's signal; a test of judges needs pools whose coverage exceeds the judge model's own accuracy. On math the judge was capped by 22.6% coverage.

The price of verification on public data. A perfect selector over random 2–5 system SWE-bench sets adds 4.8–8.3 points over the best member, at $0.062–$0.163 of extra spend per point per task. With the anchor's best measured

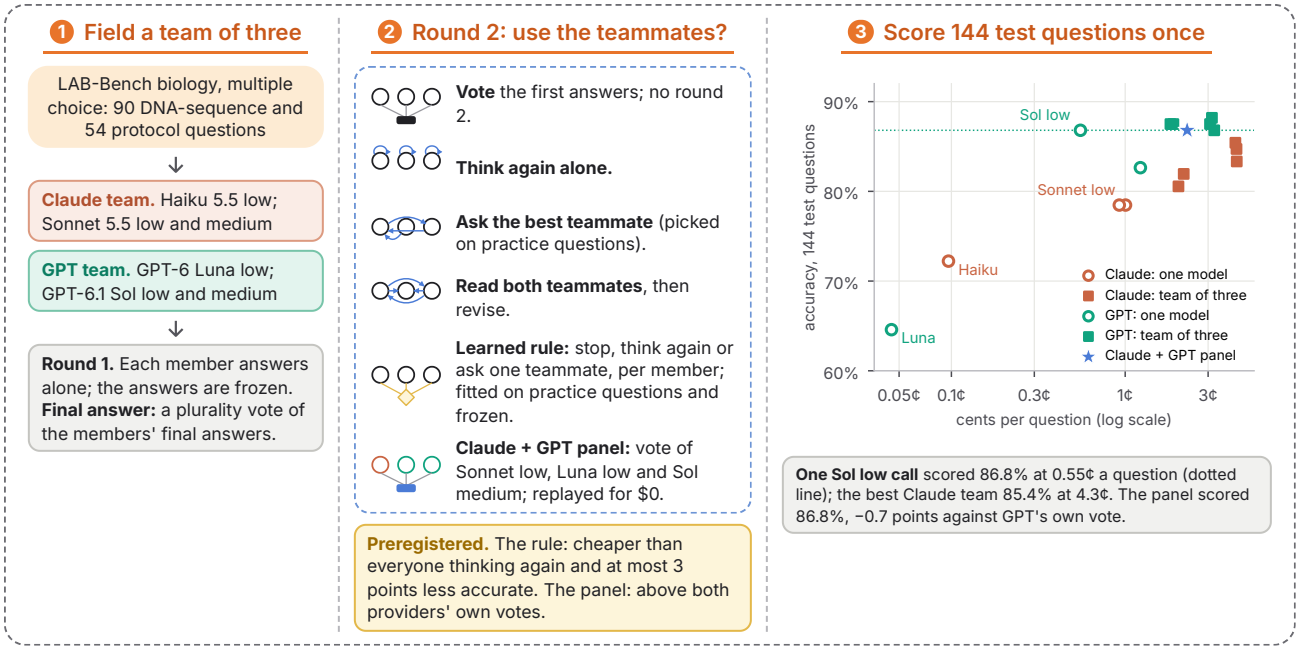


Figure 3. The LAB-Bench team study. (1) Each provider fields one cheap model and one strong model at two reasoning efforts; every member answers alone first, and the team's answer is a plurality vote of its members' final answers. (2) The round-2 methods compared: a plain vote, every member thinking again alone, asking its best teammate, reading both teammates, and a learned per-member rule, plus a panel mixing the providers that is replayed from stored calls at no cost; the rule and the panel had preregistered tests. (3) Accuracy against cents per question (log scale) on the 144 test questions, each method read once. Open circles: one model alone; squares: the team methods; star: the panel. One GPT-6.1 Sol call at low effort matches every team.

efficiency, a judge over such a set beats the most accurate single submission at no more of its dollars in at most 10% of sets (a perfect selector in at most 13%, judge calls unpriced). On MathArena the room is 1.0–4.5 points because the top systems are close to perfect.

Self-checks and execution. Sol accepted 97% of its own anchor answers (AUROC 0.69); Haiku 66% (AUROC 0.78). On SWE-bench the step count of a run ranks its own successes with median AUROC 0.68; a cascade that escalates on step count never escalates once its threshold is fitted. Execution-based checks on code are part of the open-weight testbed.

7. Different members and coordination rules

Different members. If copies fail together, members that fail differently should help (Figure 3). On 144 held-out LAB-Bench questions (DNA-sequence manipulation and protocol troubleshooting), each provider fielded one cheap model and one strong model at two reasoning efforts: Claude Haiku 5.5 with Sonnet 5.5 at low and medium effort (two models, three settings), and GPT-6 Luna with GPT-6.1 Sol at low and medium. With every member reading both teammates' answers before revising, the Claude team scored 84.7% against 78.5% for Sonnet low alone, +6.3 points (+2.1 to +11.1; POST HOC) at 4.4× the cost; a plain vote of the three first answers gained +2.1 (−0.7 to +5.6). Sol low alone scored 86.8% at $0.006 per question, and no GPT team beat it by a resolved margin (every member reading both teammates: +1.4, −2.8 to +5.6). The best Claude team (85.4% at $0.043) stayed below that one Sol call. A preregistered panel mixing the providers, chosen on practice questions and replayed from stored calls at no cost, scored 86.8% at $0.023: −0.7 points against GPT's own vote (−4.9 to +3.5) and +6.3 against Claude's (+0.7 to +11.8). The claim needed both to be positive and is not supported (CONFIRMATORY). The two providers refused disjoint questions, too few to matter for a vote.

A learned rule for when to ask a teammate. After its first answer each member could stop, think again alone, or ask one teammate. A per-member rule was fitted on practice questions, frozen, and read once; it had to be cheaper than every member thinking again, with the lower 95% bound of its accuracy difference above −3 points. Neither team's rule passed: Claude −1.4 (−6.3 to +2.8) at $0.022 against $0.044 per question, and GPT +0.7 (−3.5 to +4.9) (CONFIRMATORY). Claude's rule did worse than random choices of the same number of calls (−2.2, −4.8 to −0.1), and GPT's mostly reproduced the plain vote.

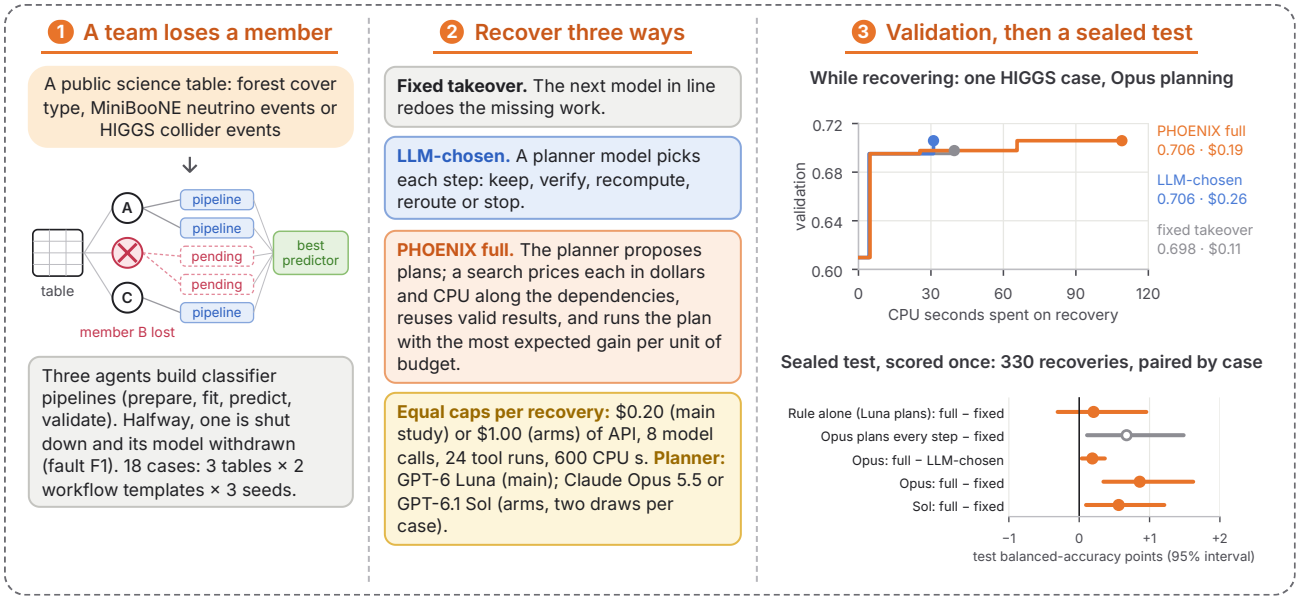


Figure 4. PHOENIX, recovery of a lost team member. (1) Three agents build classifier pipelines on a public science table; halfway, one is shut down and its model withdrawn. (2) Three recoveries run under the same caps per recovery: fixed takeover, a planner model choosing every step, and PHOENIX, which ranks candidate plans by expected gain per unit of budget along the workflow's dependencies. (3) Top: one case's validation score (training-internal rows) as the recovery spends CPU time, with each branch's final score and API dollars. Bottom: the sealed test, scored once, in balanced-accuracy points paired by case (each arm case is the mean of two draws); orange rows are preregistered contrasts, the grey row is post hoc. The rule alone is not resolved; most of the gain over takeover comes from the stronger planner.

Recovering a lost member (PHOENIX; Figure 4). Three agents build a classifier on a public science table (forest cover type, MiniBooNE neutrino events or HIGGS collisions), and halfway through one member is shut down and its model withdrawn. Three recoveries ran under the same caps on dollars, calls, tools and CPU time: ordinary takeover by the next model, a planner model choosing every step, and PHOENIX, which ranks candidate plans by expected gain per dollar along the workflow's dependencies. Every recovered predictor was sealed and then scored once on a held-out split (330 recoveries, balanced-accuracy points). Where the rule can be isolated, with one cheap model doing everything, PHOENIX minus takeover is +0.20 (−0.31 to +0.95; CONFIRMATORY, not resolved). With Claude Opus 5.5 planning it beats takeover by +0.86 (+0.34 to +1.62), but Opus choosing every step without the rule already gains +0.67 (+0.11 to +1.48; POST HOC); the rule adds +0.19 (+0.04 to +0.36) at $0.126 against $0.231 per recovery. With GPT-6.1 Sol planning the gain over takeover is +0.56 (+0.10 to +1.21) and fragile, and two draws of the same method on one case differed by up to 5.85 points. The planner model mattered; the rule's own value is not shown.

8. Choosing members per budget

Because every call is stored, every team in this paper is rescored by replay at no cost, and the best team for each dollar and latency budget can be chosen from calls already made (Figure 5). On the anchor's BBEH questions 42 budget cells have a feasible team, and in every one the winner is a single model: one call in 24 cells and a vote of its own copies in 18. Chosen on half the questions and scored on the other half, the choice loses 0.9–1.6 points on average against the best team for the held-out half (EXPLORATORY). This is stability within one benchmark, not transfer to a new task.

9. The open-weight testbed (design; results pending)

The testbed varies precision (BF16, W8, W4, W3), size (Qwen3.5 2B, 4B, 9B) and thinking budget (off, 1k, 4k) with several samples per item on 363 items (200 BBEH, 63 competition math, 100 LiveCodeBench), so width, depth and verification can be bought at measured GPU cost. Its preregistration (T1–T5) has power by simulation of 0.94 (T1: the W4 loss survives voting on hard items), 1.00 (T2), 1.00 (T3: depth against cost-matched width), 1.00 (T4: judge coupling to difficulty) and 0.65 (T5, exploratory: compositional prediction) (DESIGN). The pilot is estimated at $4.1 of GPU time. No GPU has run yet; no result is reported.

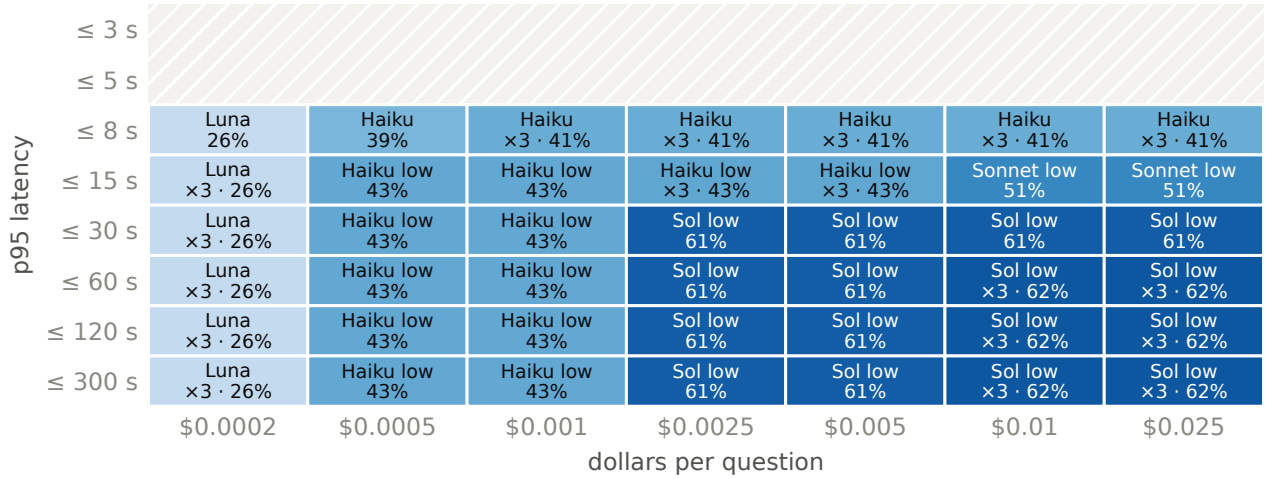


Figure 5. The budget map on the anchor's BBEH questions: the most accurate team within each dollar budget (columns) and 95th-percentile latency budget (rows), chosen by replaying stored calls. Shading is its accuracy; "×3" marks a vote of three calls of one model; hatched rows have no team fast enough. Every winner is a single model.

10. Related work

Shared failure and its ceiling. Chen (2026) show that any routing, voting or cascade policy returning one model's answer is bounded by one minus the share of queries every model fails, certify that share from a graded set, and find that a one-factor copula underprices it on open-ended mathematics; they exclude thinking variants and do not hold dollars fixed. We take the ceiling as given and ask how depth moves it and at what price. Kim et al. (2025) measure correlated errors across hundreds of models without an item-difficulty term; Turkmen et al. (2026) prove an error floor under a Gaussian latent factor; Denisov-Blanch et al. (2026) find that polling does not improve truthfulness because errors are shared. Ge et al. (2026) fit an ability-minus-difficulty model to agentic coding leaderboards to predict tasks and agents; we use the same scale for composition.

Width. Chen et al. (2024) show that vote accuracy can rise and then fall with more calls of one model on a mix of easy and hard queries; they count calls, not dollars. Chen et al. (2025) assign different models to the modules of a compound system with a labelled search, which can beat any single model; this is role allocation, not parallel width.

Verification and cascades. Singhi et al. (2025) find self-consistency more compute-efficient than generative verification at most practical budgets, with verification ahead only at many times the compute; our comparator is the judge model's own answer at measured dollars. Conformal cascades (Dou et al., 2026) need labelled calibration data; C3PO (Valkanas et al., 2025) anchors a label-free cascade on the strongest model's answers and so inherits its errors. Learn-then-Test (Angelopoulos et al., 2021) gives the risk control we plan for a runtime router.

11. Limitations

Models and prices are those of October 2026, and public leaderboards mix scaffolds and dates. Most Atlas results are post hoc and uncorrected for the many comparisons made. Contamination cannot be ruled out: BBEH has been public since 2025, so "fresh" means new to this project, and LiveCodeBench predates the testbed's models. The API studies are small (533 items in the anchor) and use two vendors; the AH2 decision covers the first effort step for cheap models on these items only; beyond it the depth study describes flattening gains but does not test depth against width. The anchor's judge test was capped by its pools. The team studies are small (144 LAB-Bench test questions, 18 PHOENIX cases on tables near their accuracy ceiling), several team comparisons are post hoc, and the budget map's stability is measured within one benchmark. Answer format moves the difficulty axis (Chen, 2026); our anchor tasks are free-form. The all-wrong tail on small math sets is not resolved. The anchor's systematic shares have no interval until the gold files are restored.

12. Conclusion and next steps

Which members a system calls matters more than how it organizes them. On the data we have, failures are mostly shared, and a factor model with one free difficulty per item prices the shared tail on code better than a one-factor copula given the same data. Reasoning moves a system along the shared difficulty axis without changing which items it

fails. The first reasoning step of cheap models beat a dollar-matched vote of their no-reasoning calls on sealed items; whether depth beats width beyond that step is open, and our public pairs predict about zero at the top of a model's range. Selection by agreement did not beat the best system's own vote at matched calls, and a test of judges still needs pools that cover more than the judge answers alone. Different members helped only where the best member left room, and smarter coordination rules, for asking a teammate or for recovering a lost member, showed no gain we can credit to the rule. A preregistered handoff test, how much of a strong model's accuracy a cheaper team completes from the strong model's partial reasoning, has run and is not yet read. The next steps (roadmap in research/v4/roadmap.md): 2,000+ items across 8 or more task families including agentic ones, with a time-split replication; a third vendor used once on the frozen design; the open-weight testbed; batch and cache pricing; and a budget allocator that turns a 30–50 item pilot and a price table into a depth, width and verification mix with predicted accuracy, cost and uncertainty.

References

Angelopoulos, A. N., Bates, S., Candès, E. J., Jordan, M. I., and Lei, L. Learn then test: Calibrating predictive algorithms to achieve risk control. *arXiv preprint arXiv:2110.01052*, 2021.

Balunović, M., Dekoninck, J., Petrov, I., Jovanović, N., and Vechev, M. MathArena: Evaluating LLMs on uncontaminated math competitions. *arXiv preprint arXiv:2505.23281*, 2025.

Chen, J. When does combining language models help? a co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models. *arXiv preprint arXiv:2606.27288*, 2026.

Chen, L., Davis, J. Q., Hanin, B., Bailis, P., Stoica, I., Zaharia, M., and Zou, J. Are more LLM calls all you need? towards scaling laws of compound inference systems. *arXiv preprint arXiv:2403.02419*, 2024.

Chen, L., Davis, J. Q., Hanin, B., Bailis, P., Zaharia, M., Zou, J., and Stoica, I. Optimizing model selection for compound AI systems. *arXiv preprint arXiv:2502.14815*, 2025.

Dawid, A. P. and Skene, A. M. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Journal of the Royal Statistical Society, Series C*, 28(1): 20–28, 1979.

Denisov-Blanch, Y., Kazdan, J., Chudnovsky, J., Schaeffer, R., Guan, S., Adeshina, S., and Koyejo, S. Consensus is not verification: Why crowd wisdom strategies fail for LLM truthfulness. *arXiv preprint arXiv:2603.06612*, 2026.

Dou, Y., Lian, S., and Li, S. Conformal cascade: Distribution-free accuracy guarantees for multi-tier LLM inference. *arXiv preprint arXiv:2607.25018*, 2026.

Ge, C., Kryvosheieva, D., Fried, D., Girit, U., and Hariharan, K. Agent psychometrics: Task-level performance prediction in agentic coding benchmarks. *arXiv preprint arXiv:2604.00594*, 2026.

Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., and Narasimhan, K. SWE-bench: Can language models resolve real-world GitHub issues? In *International Conference on Learning Representations (ICLR)*, 2024.

Kazemi, M. et al. BIG-Bench Extra Hard. *arXiv preprint arXiv:2502.19187*, 2025.

Kim, E., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2506.07962.

Singhi, N., Bansal, H., Hosseini, A., Grover, A., Chang, K.-W., Rohrbach, M., and Rohrbach, A. When to solve, when to verify: Compute-optimal problem solving and generative verification for LLM reasoning. In *Conference on Language Modeling (COLM)*, 2025. arXiv:2504.01005.

Turkmen, Y., Buyukates, B., and Bastopcu, M. Don't always pick the highest-performing model: An information theoretic view of LLM ensemble selection. *arXiv preprint arXiv:2602.08003*, 2026.

Valkanas, A., Pal, S., Rumiantsev, P., Zhang, Y., and Coates, M. C3PO: Optimized large language model cascades with probabilistic cost constraints for reasoning. *arXiv preprint arXiv:2511.07396*, 2025.

White, C. et al. LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint arXiv:2406.19314*, 2024.

A. Reproducibility

Preregistration. Every paid run carried a committed `prereg:` block (question, prediction, kill rule, decision) that code checked before the run could spend; the full texts are in the repository's research/ folder. Tests whose simulated power at the chosen n fell below 0.70 were reported as exploratory.

Audits. An independent audit read every design and every paid run's outcome before the next step: 135 audits, with 32 GO, 99 ADJUST and 2 STOP verdicts; every ADJUST was resolved before further spend.

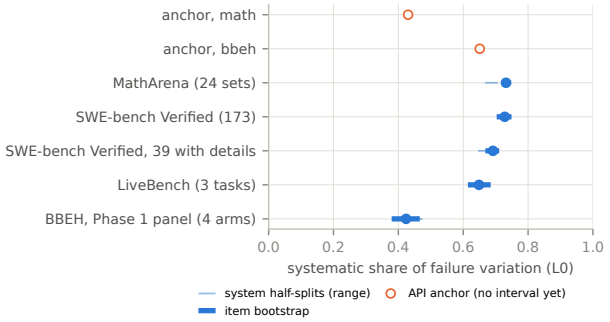


Figure 6. Systematic failure share per source (L0): item-bootstrap 95% interval (thick) and the range over system half-splits (thin; each half fitted on its own). The anchor's shares (hollow) await the gold files for an interval.

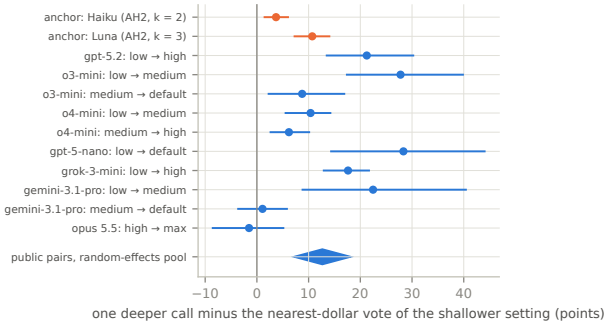


Figure 7. One deeper call minus the nearest-dollar vote of the shallower setting: the anchor's confirmatory AH2 (orange, fresh sealed items, the first effort step of two cheap models) and public MathArena pairs (blue, post hoc) with item-bootstrap 95% intervals and their random-effects pool.

Cost. Total API spend $196.64 over 502 runs, under hard per-account caps enforced in code. Every call is stored, so every team in this paper is rescored by replay at no cost. The per-run ledger and the status of every number in the paper are generated in the repository (paper/runs_table.tex, paper/numbers_table.tex).

B. Additional figures

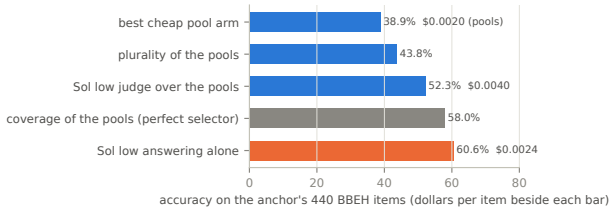


Figure 8. The anchor's BBEH verification read: a judge over cheap pools against the judge model answering alone. The pools' coverage (grey) is below the judge model's own accuracy, which caps every selector over them.

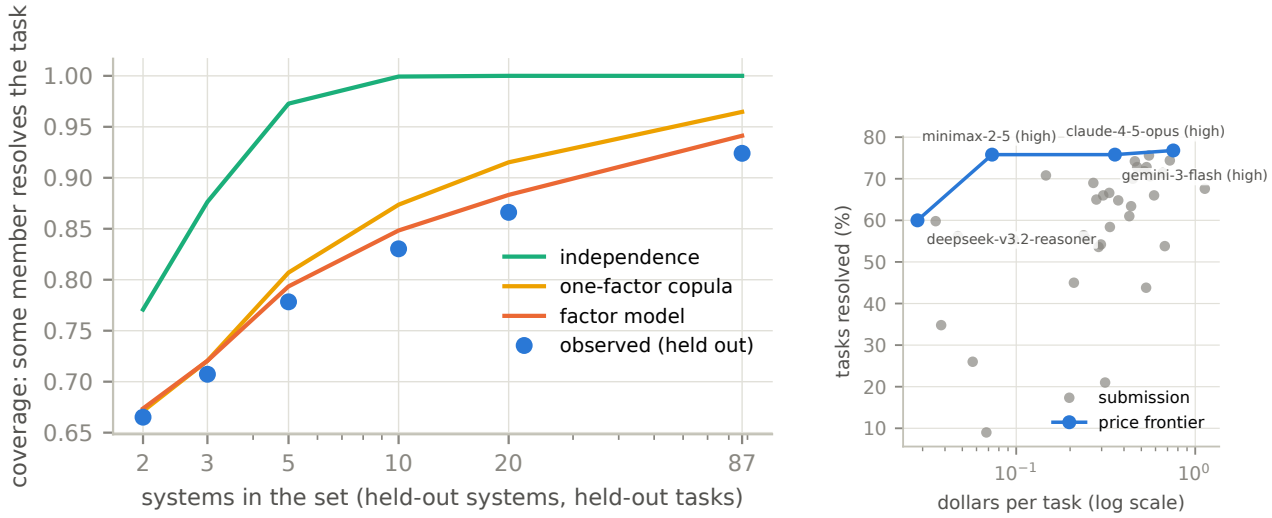


Figure 9. Left: coverage of sets of held-out SWE-bench Verified systems on held-out tasks, observed and predicted from the calibration tasks only (P8). Right: tasks resolved against dollars per task, SWE-bench Verified submissions with costs; the price frontier in blue (A7).

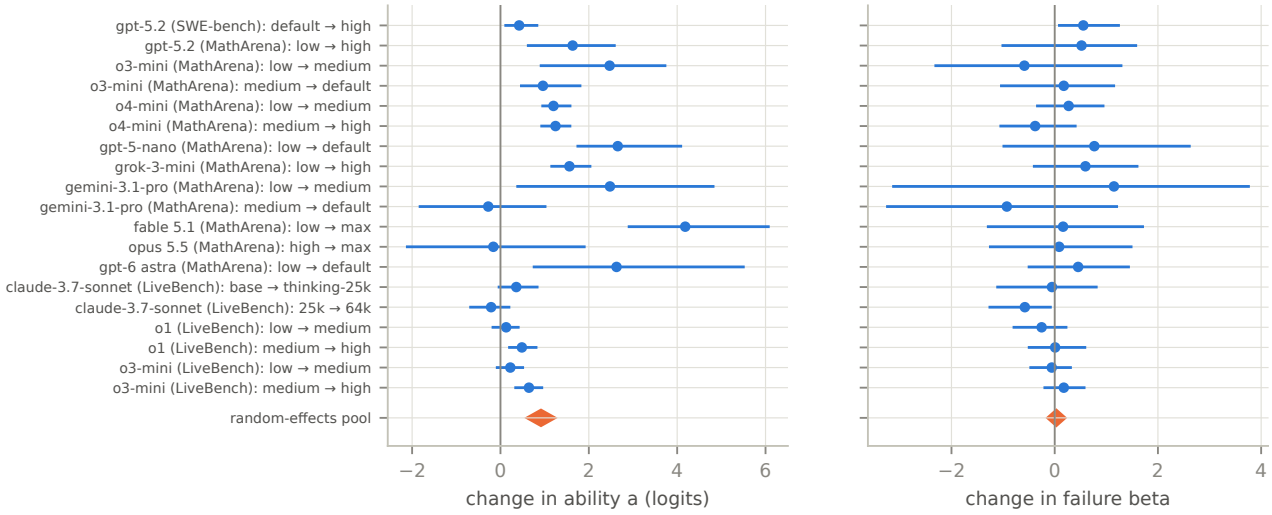


Figure 10. One model at two adjacent reasoning settings (P2): change in ability a and in failure slope β with item-bootstrap 95% intervals, and the DerSimonian–Laird random-effects pool (diamond).

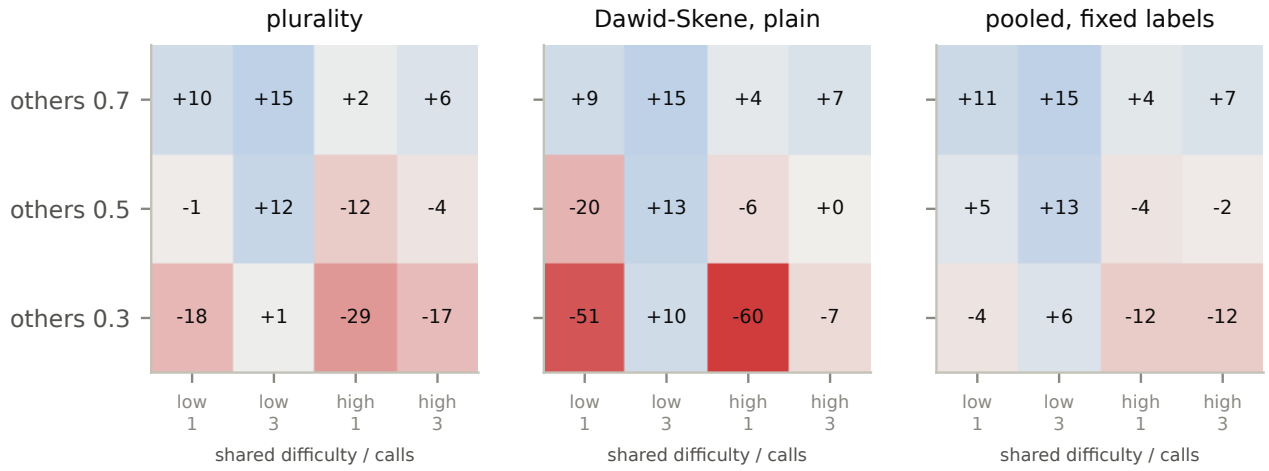


Figure 11. Label-free selection, simulation map (P4): gain over the best system (points) when one system at 0.85 sits among others at 0.3, 0.5 or 0.7, at low or high shared difficulty (systematic share 0.07 or 0.55), with one or three calls per system; averaged over 3, 5 and 9 systems.